WORKED SOLUTION

STEP II 2023, Question 9

Mechanics • Connected Particles on Slopes

STEP II 2023, Question 9

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

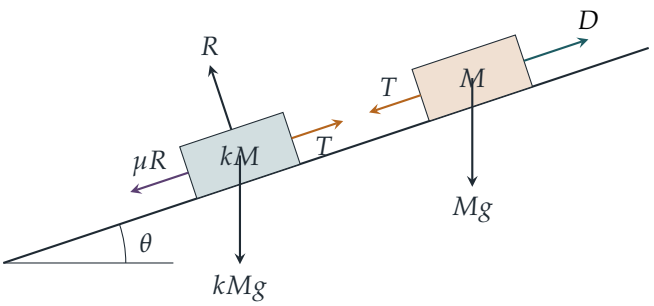
The key idea

Write down Newton's second law separately for the truck and for the trailer in each of the three situations. Going down the slope is just going up a slope of angle $-\alpha$, so one set of equations covers all three cases. The truck's equation alone gives the relation in (i) almost for free. For (ii), the whole problem comes down to comparing $1 - \cos\alpha$ with $\sin\alpha$.

Solution

We model the truck and trailer as particles joined by a light rigid bar along the line of motion. Both move in the direction of the truck, so the trailer's friction acts against the motion with magnitude μR, where R is the normal reaction on the trailer. The tension T pulls the trailer forwards and the truck backwards. (If T came out negative, the bar would be in thrust; the algebra is unchanged.)

Treat the slope as having angle θ, measured so that $\theta = \alpha$ is uphill, $\theta = 0$ is level and $\theta = -\alpha$ is downhill.



With acceleration a up the slope, resolving along the slope for each body, and perpen-

dicular to it for the trailer ($R = kMg\cos\theta$), gives

$$\text{truck:} \qquad D - T - Mg\sin\theta = Ma, \tag{1}$$

$$\text{trailer:} \qquad T - kMg\sin\theta - \mu kMg\cos\theta = kMa. \tag{2}$$

Adding (1) and (2) eliminates T:

$$a = \frac{D}{(1+k)M} - g\sin\theta - \frac{\mu k g\cos\theta}{1+k}.$$

Substituting into (1), $T = D - Mg\sin\theta - Ma$, so

$$T = D - \frac{D}{1+k} + \frac{\mu kMg\cos\theta}{1+k} = \frac{kD + \mu kMg\cos\theta}{1+k}.$$

The $\sin\theta$ terms cancel, so the tension depends on θ only through $\cos\theta$.

Part (i)

Since $\cos(-\alpha) = \cos\alpha$, the cases $\theta = \alpha$ and $\theta = -\alpha$ give the same tension:

$$T_1 = T_3 = \frac{kD + \mu kMg\cos\alpha}{1+k}, \qquad T_2 = \frac{kD + \mu kMg}{1+k}.$$

For the second result we only need the truck equation (1) in the three cases:

$$\begin{aligned} Ma_1 &= D - T_1 - Mg\sin\alpha, \\ Ma_2 &= D - T_2, \\ Ma_3 &= D - T_3 + Mg\sin\alpha. \end{aligned}$$

Adding the first and third equations and using $T_3 = T_1$ gives $M(a_1 + a_3) = 2D - 2T_1$. Subtracting twice the middle equation,

$$M(a_1 + a_3 - 2a_2) = (2D - 2T_1) - (2D - 2T_2) = 2(T_2 - T_1),$$

as required.

Part (ii)(a)

First inequality. By (i), $a_2 < \frac{1}{2}(a_1 + a_3)$ is equivalent to $T_2 > T_1$ (since $M > 0$). From the formulas above,

$$T_2 - T_1 = \frac{\mu kMg\,(1 - \cos\alpha)}{1+k} > 0,$$

because $0 < \alpha < \frac{1}{2}\pi$ gives $\cos\alpha < 1$, and $\mu > 0$ since friction acts. So $a_2 < \frac{1}{2}(a_1 + a_3)$.

Second inequality. $\frac{1}{2}(a_1 + a_3) < a_3$ is equivalent to $a_1 < a_3$. From the formula for a, the D and friction terms are the same for $\theta = \pm\alpha$, so

$$a_3 - a_1 = 2g\sin\alpha > 0.$$

(Equivalently, subtract the truck equations: $M(a_3 - a_1) = 2Mg\sin\alpha$, using $T_1 = T_3$.) Hence

$$a_2 < \tfrac{1}{2}(a_1 + a_3) < a_3.$$

Part (ii)(b)

Using the formula for a with $\theta = 0$ and $\theta = \alpha$,

$$a_2 - a_1 = g\sin\alpha - \frac{\mu k g\,(1-\cos\alpha)}{1+k} = g\left[\sin\alpha - \frac{\mu k}{1+k}(1-\cos\alpha)\right].$$

Ask yourself Going uphill costs $g\sin\alpha$ of acceleration, but the reduced normal reaction saves some friction. Which effect wins?

The friction saving is the smaller one. First, $\dfrac{k}{1+k} < 1$ and $\mu < 1$, so $0 \le \dfrac{\mu k}{1+k} < 1$. Second, using the half-angle formulae,

$$1-\cos\alpha = 2\sin^2\tfrac{1}{2}\alpha, \qquad \sin\alpha = 2\sin\tfrac{1}{2}\alpha\cos\tfrac{1}{2}\alpha,$$

so $\dfrac{1-\cos\alpha}{\sin\alpha} = \tan\tfrac{1}{2}\alpha < 1$, since $0 < \tfrac{1}{2}\alpha < \tfrac{1}{4}\pi$. Therefore

$$\frac{\mu k}{1+k}(1-\cos\alpha) < 1-\cos\alpha < \sin\alpha,$$

where the first inequality is strict because $1-\cos\alpha > 0$. So $a_2 - a_1 > 0$, that is, $a_1 < a_2$. This is where $\mu < 1$ is used. With a very heavy trailer, a large μ and a steep slope, the friction saving can exceed $g\sin\alpha$.

Checks

- Take $M = 1000\,\text{kg}$, $k = 0.5$, $\mu = 0.4$, $\alpha = 30^\circ$, $D = 5000\,\text{N}$ and $g = 9.8\,\text{m s}^{-2}$. Then $a_1 \approx -2.698$, $a_2 \approx 2.027$ and $a_3 \approx 7.102\,\text{m s}^{-2}$, with $T_1 = T_3 \approx 2798\,\text{N}$ and $T_2 \approx 2973\,\text{N}$. Indeed $\frac{1}{2}(a_1 + a_3) \approx 2.202$ lies strictly between a_2 and a_3, and $a_1 < a_2$. A computer check over many random parameter sets with $\mu < 1$ confirmed all the identities and inequalities.
- If $\mu = 0$, the tension is $kD/(1+k)$ on every slope and a_2 is exactly the mean of a_1 and a_3. That is why (ii)(a) needs friction to be present.
- With $\mu = 3$, $k = 100$ and $\alpha = 1.5$ radians, $a_1 > a_2$, so the condition $\mu < 1$ in (b) really matters.

Common slips

- Using $\mu k M g$ for the friction on the slopes. The normal reaction there is $kMg\cos\alpha$, and this difference is the whole content of part (ii).
- Getting the sign of the weight component wrong when going downhill. Writing everything in terms of θ and then setting $\theta = -\alpha$ avoids this.
- In (ii)(b), arguing only that $\mu k/(1+k) < 1$ without comparing $1-\cos\alpha$ with $\sin\alpha$. Both facts are needed.

Beyond STEP

Splitting a system into bodies and applying Newton's laws to each, with internal forces appearing in equal and opposite pairs, is the basic method of rigid-body and multi-particle mechanics. Adding the equations gives the motion of the centre of mass, and the internal force drops out. The inequality $1 - \cos\alpha < \sin\alpha$ is also a small example of comparing functions using half-angle identities, a technique that is useful throughout analysis. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops these ideas carefully.

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